Cruise Control Subsystem - AUTOSAR High-Level Design

Version 1.1

1 Introduction

This document describes the high-level design of the cruise control subsystem on the vehicle ECU. Intended readers are system architects, software developers, integration engineers and test engineers.

2 Architecture overview

The subsystem consists of 6 software components in the application layer. Components communicate through ports typed by port interfaces, connected by the RTE.

2.1 Component catalogue

Component	Kind	Runnable	Period (ms)
SWC_WheelSpeedSensor	Sensor-actuator SWC	Rn_ReadWheelSpeed	5
SWC_VehicleSpeedCalc	Application SWC	Rn_CalcVehicleSpeed	10
SWC_CruiseControl	Application SWC	Rn_CruiseControlStep	10
SWC_EngineControl	Application SWC	Rn_ApplyThrottle	10
SWC_DisplayCluster	Application SWC	Rn_UpdateDisplay	100
SWC_BrakeManager	Sensor-actuator SWC	Rn_ReadBrake	10

3 Software component descriptions

3.1 SWC_WheelSpeedSensor

Reads the wheel speed pulses through the ECU abstraction layer and publishes the filtered wheel speed to the application layer.

Ports

Port	Dir	Interface	Interface type	Data element	Data type	Unit
Pp_WheelSpeed	P	If_WheelSpeed	Sender-Receiver	wheelSpeed	uint16	km/h

Functional flow

Runnable Rn_ReadWheelSpeed is triggered every 5 ms by a timing event. It writes wheelSpeed through Pp_WheelSpeed.

3.2 SWC_VehicleSpeedCalc

Calculates the vehicle speed and the longitudinal acceleration from the wheel speed and publishes both values.

Ports

Port	Dir	Interface	Interface type	Data element	Data type	Unit
Rp_WheelSpeed	R	If_WheelSpeed	Sender-Receiver	wheelSpeed	uint16	km/h
Pp_VehicleSpeed	P	If_VehicleSpeed	Sender-Receiver	vehicleSpeed	uint16	km/h
Pp_VehicleAccel	P	If_VehicleAccel	Sender-Receiver	vehicleAccel	sint16	m/s2

Functional flow

Runnable Rn_CalcVehicleSpeed is triggered every 10 ms by a timing event. It reads wheelSpeed through Rp_WheelSpeed. It writes vehicleSpeed through Pp_VehicleSpeed. It writes vehicleAccel through Pp_VehicleAccel.

3.3 SWC_CruiseControl

Holds the driver set speed. It consumes the vehicle speed from the Vehicle Speed Calculator (SWC_VehSpeedCalc) and the brake status, and requests throttle from the engine control.

Ports

Port	Dir	Interface	Interface type	Data element	Data type	Unit
Rp_VehicleSpeed	R	If_VehicleSpeed	Sender-Receiver	vehicleSpeed	uint16	km/h
Rp_BrakeStatus	R	If_BrakeStatus	Sender-Receiver	brakeActive	boolean	-
Pp_ThrottleRequest	P	If_ThrottleRequest	Sender-Receiver	throttlePct	uint8	%

Functional flow

Runnable Rn_CruiseControlStep is triggered every 10 ms by a timing event. It reads vehicleSpeed through Rp_VehicleSpeed. It reads brakeActive through Rp_BrakeStatus. It writes throttlePct through Pp_ThrottleRequest.

3.4 SWC_EngineControl

Converts the throttle request into an actuator command for the electronic throttle.

Ports

Port	Dir	Interface	Interface type	Data element	Data type	Unit
Rp_ThrottleRequest	R	If_ThrottleRequest	Client-Server	throttlePct	uint8	%

Functional flow

Runnable Rn_ApplyThrottle is triggered every 10 ms by a timing event. It reads throttlePct through Rp_ThrottleRequest.

3.5 SWC_DisplayCluster

Shows the current vehicle speed to the driver on the instrument cluster.

Ports

Port	Dir	Interface	Interface type	Data element	Data type	Unit
Rp_VehicleSpeed	R	If_VehicleSpeed	Sender-Receiver	vehicleSpeed	uint8	mph

Functional flow

Runnable Rn_UpdateDisplay is triggered every 100 ms by a timing event. It reads vehicleSpeed through Rp_VehicleSpeed.

3.6 SWC_BrakeManager

Reads the brake pedal switch and publishes the brake status.

Ports

Port	Dir	Interface	Interface type	Data element	Data type	Unit
Pp_BrakeStatus	P	If_BrakeStatus	Sender-Receiver	brakeActive	boolean	-

Functional flow

Runnable Rn_ReadBrake is triggered every 10 ms by a timing event. It writes brakeActive through Pp_BrakeStatus.

4 Revision history

Version	Date	Change
1.0	2026-09-01	Initial release
1.1	2026-09-20	Added SWC_BrakeManager; periods of SWC_WheelSpeedSensor and SWC_CruiseControl changed